

1. (a)  $\int_1^\infty \frac{1}{1+x^2} dx < \int_1^\infty \frac{1}{x^2} dx = \left[-\frac{1}{x}\right]_1^\infty = 1$   
 $\Rightarrow \int_1^\infty \frac{1}{1+x^2} dx = [\tan^{-1} x]_1^\infty = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$
- (b)  $\int_2^\infty \frac{dx}{x^2-1} = \lim_{b \rightarrow \infty} \int_2^b \frac{dx}{x^2-1} = \lim_{b \rightarrow \infty} \left[ \frac{1}{2} \ln \left| \frac{x-1}{x+1} \right| \right]_2^b$   
 $= \lim_{b \rightarrow \infty} \left[ \frac{1}{2} \ln \left( \frac{b-1}{b+1} + \frac{1}{2} \ln 3 \right) \right] = \frac{1}{2} \ln 3$
- (c) Let  $u = \ln x \Rightarrow du = \frac{1}{x} dx$   
Then  $\int \frac{dx}{x(\ln x)^2} = \int \frac{du}{u^2} = -\frac{1}{u} + C = -\frac{1}{\ln x} + C$
- (d)  $\int_0^\infty e^{-x} \sin x dx = \lim_{b \rightarrow \infty} \int_0^b e^{-x} \sin x dx = \lim_{b \rightarrow \infty} -\frac{1}{2} [e^{-x} \cos x + e^{-x} \sin x]_0^b$   
 $= \lim_{b \rightarrow \infty} \frac{1}{2} [1 - e^{-b} \cos b - e^{-b} \sin b] = \frac{1}{2}$
- (e)  $\int_{-\infty}^0 x e^x dx = \lim_{a \rightarrow -\infty} \int_a^0 x e^x dx = \lim_{a \rightarrow -\infty} [x e^x - e^x]_a^0 = \lim_{a \rightarrow -\infty} [-1 - a e^a + e^a] = -1$
- (f) Since  $\int_1^2 \frac{1}{x^2-5x+6} dx = \lim_{b \rightarrow 2^-} \int_1^b \frac{dx}{(x-2)(x-3)}$   
 $= \lim_{b \rightarrow 2^-} \left[ \ln \left| \frac{x-3}{x-2} \right| \right]_1^b = \lim_{b \rightarrow 2^-} \left( \ln \left| \frac{b-3}{b-2} \right| - \ln 2 \right)$  diverges  
So  $\int_1^4 \frac{1}{x^2-5x+6} dx$  diverges as well.
- (g) Since  $\int_0^\infty \frac{1}{e^x + e^{-x}} dx = \lim_{b \rightarrow \infty} \int_0^b \frac{1}{e^x + e^{-x}} dx = \lim_{b \rightarrow \infty} [\tan^{-1} e^x]_0^b = \frac{\pi}{2} - \frac{\pi}{4}$   
and  $\int_{-\infty}^0 \frac{1}{e^x + e^{-x}} dx = \lim_{b \rightarrow -\infty} \int_b^0 \frac{1}{e^x + e^{-x}} dx = \lim_{b \rightarrow -\infty} [\tan^{-1} e^x]_b^0 = \frac{\pi}{4}$   
So  $\int_{-\infty}^\infty \frac{1}{e^x + e^{-x}} dx = \int_0^\infty \frac{1}{e^x + e^{-x}} dx + \int_{-\infty}^0 \frac{1}{e^x + e^{-x}} dx = \frac{\pi}{2}$
- (h)  $\int_0^1 \arcsin x dx = [x \sin^{-1} x]_0^1 - \int_0^1 \frac{x}{\sqrt{1-x^2}} dx = \frac{\pi}{2} - \lim_{a \rightarrow 1^-} \int_0^a \frac{x}{\sqrt{1-x^2}} dx$   
Let  $u = 1 - x^2 \Rightarrow du = -2x dx$   
and  $x = 0 \Rightarrow u = 1$ ;  $x = a \Rightarrow u = 1 - a^2$   
Then  $\int_0^a \frac{x}{\sqrt{1-x^2}} dx = -\frac{1}{2} \int_1^{1-a^2} \frac{1}{\sqrt{u}} du = [-\sqrt{u}]_1^{1-a^2} = 1 - \sqrt{1-a^2}$   
Thus,  $\int_0^1 \arcsin x dx = \frac{\pi}{2} - \lim_{a \rightarrow 1^-} (1 - \sqrt{1-a^2}) = \frac{\pi}{2} - 1$
- (i) Let  $I = \int_1^\infty 2^{-x^2} dx = \int_1^\infty 2^{-y^2} dy$   
Then  $I^2 = \int_1^\infty \int_1^\infty 2^{-(x^2+y^2)} dx dy$   
Let  $x = r \cos \theta$ ;  $y = r \sin \theta \Rightarrow r^2 = x^2 + y^2$   
So  $I^2 = 2 \int_0^{\frac{\pi}{4}} \int_{\csc \theta}^\infty 2^{-r^2} r dr d\theta = \int_0^{\frac{\pi}{4}} \int_{\csc \theta}^\infty 2^{-r^2} dr^2 d\theta = \int_0^{\frac{\pi}{4}} \left[ \frac{-1}{\ln 2} e^{-\ln 2 \cdot \csc^2 \theta} \right] d\theta$   
 $\Rightarrow I \doteq 0.2544$

- (j)  $\int_e^\infty \frac{dx}{\sqrt{x+1} \ln x}$  diverges by comparison with  $\int_e^\infty \frac{dx}{\sqrt{x+1} \ln(x+1)}$
- (k)  $\int_0^\infty \cos^2 x dx = \lim_{b \rightarrow \infty} \left[ \frac{x}{2} + \frac{\sin 2x}{4} \right]_0^b = \lim_{b \rightarrow \infty} \left( \frac{b}{2} + \frac{\sin 2b}{4} \right) = \infty$   
 So  $\int_0^\infty \cos^2 x dx$  diverges.
- (l)  $\int_0^{\frac{\pi}{2}} \frac{\cos x}{\sqrt{\sin x}} dx = \lim_{a \rightarrow 0^+} \int_a^{\frac{\pi}{2}} \frac{\cos x}{\sqrt{\sin x}} dx = \lim_{a \rightarrow 0^+} \left[ 2\sqrt{\sin x} \right]_a^{\frac{\pi}{2}} = 2$
2. (a)  $\int_0^\infty x^r e^{-x} dx$  diverges if  $r \leq -1$   
 Since  $\int_0^\infty x^r e^{-x} dx = \int_0^1 x^r e^{-x} dx + \int_1^\infty x^r e^{-x} dx$  and  $\int_0^1 x^r e^{-x} dx$  diverges  
 For any  $r > -1$ , we can find  $k$  such that  $x^r < e^{x/2}$  for  $x \geq k$ . ( $e^{x/2}$  grows faster than any other power of  $x$ )  
 Then  $\int_0^\infty x^r e^{-x} dx < \int_0^k x^r e^{-x} dx + \int_k^\infty e^{-x/2} dx$ , which converges.  
 Thus,  $\int_0^\infty x^r e^{-x} dx$  converges for all  $r > -1$ .
- (b) For  $n = 1$ :  $\int_0^\infty x e^{-x} dx = \lim_{b \rightarrow \infty} [-x e^{-x} - e^{-x}]_0^b = \lim_{b \rightarrow \infty} [-b e^{-b} - e^{-b} + 1] = 1$   
 Suppose that the statement is true for  $n = k$ , i.e.  $\int_0^\infty x^k e^{-x} dx = k!$   
 Then for  $n = k + 1$ :  $\int_0^\infty x^{k+1} e^{-x} dx = \lim_{b \rightarrow \infty} \left( [-x^{k+1} e^{-x}]_0^b + (k+1) \int_0^b x^k e^{-x} dx \right)$   
 $= \lim_{b \rightarrow \infty} (-b^{k+1} e^{-b}) + (k+1) \int_0^\infty x^k e^{-x} dx = 0 + (k+1)k! = (k+1)!$
3.  $\int_0^\infty \frac{1}{\sqrt{x}(1+x)} dx = \int_0^1 \frac{1}{\sqrt{x}(1+x)} dx + \int_1^\infty \frac{1}{\sqrt{x}(1+x)} dx = \lim_{a \rightarrow 0^+} \int_a^1 \frac{1}{\sqrt{x}(1+x)} dx + \lim_{b \rightarrow \infty} \int_1^b \frac{1}{\sqrt{x}(1+x)} dx$   
 Let  $u = \sqrt{x}$   
 Then  $\int \frac{1}{\sqrt{x}(1+x)} dx = \int \frac{2}{1+u^2} du = 2 \arctan u + C = 2 \arctan \sqrt{x} + C$   
 Therefore,  $\lim_{a \rightarrow 0^+} \int_a^1 \frac{1}{\sqrt{x}(1+x)} dx = \lim_{a \rightarrow 0^+} [2 \arctan \sqrt{x}]_a^1 = \lim_{a \rightarrow 0^+} 2\left[\frac{\pi}{4} - \arctan \sqrt{a}\right] = \frac{\pi}{2}$   
 and  $\lim_{b \rightarrow \infty} \int_1^b \frac{1}{\sqrt{x}(1+x)} dx = \lim_{b \rightarrow \infty} [2 \arctan \sqrt{x}]_1^b = \lim_{b \rightarrow \infty} 2\left[\arctan \sqrt{b} - \frac{\pi}{4}\right] = \frac{\pi}{2}$   
 Thus,  $\int_0^\infty \frac{1}{\sqrt{x}(1+x)} dx = \pi$
4. Although  $\lim_{b \rightarrow \infty} \int_{-b}^b \sin x dx = \lim_{b \rightarrow \infty} [-\cos x]_{-b}^b = \lim_{b \rightarrow \infty} [-\cos b + \cos(-b)] = 0$   
 $\int_0^\infty \sin x dx = \lim_{b \rightarrow \infty} \int_0^b \sin x dx = \lim_{b \rightarrow \infty} [-\cos x]_0^b = 1 - \lim_{b \rightarrow \infty} \cos b \Rightarrow$  The limit does not exist.
5. For all real  $t$ ,  $-\frac{t^2}{2} < t + 1$ .  
 Therefore,  $\int_{-\infty}^x e^{-t^2/2} dt$  converges by comparison with  $\int_{-\infty}^x e^{t+1} dt$ .

$$6. \int_0^1 \arcsin x dx = [x \sin^{-1} x]_0^1 - \int_0^1 \frac{x}{\sqrt{1-x^2}} dx = \frac{\pi}{2} - \lim_{a \rightarrow 1^-} \int_0^a \frac{x}{\sqrt{1-x^2}} dx$$

$$\text{Let } u = 1 - x^2 \Rightarrow du = -2x dx$$

$$\text{and } x = 0 \Rightarrow u = 1 ; x = a \Rightarrow u = 1 - a^2$$

$$\text{Then } \int_0^a \frac{x}{\sqrt{1-x^2}} dx = -\frac{1}{2} \int_1^{1-a^2} \frac{1}{\sqrt{u}} = [-\sqrt{u}]_1^{1-a^2} = 1 - \sqrt{1-a^2}$$

$$\text{Thus, } \int_0^1 \arcsin x dx = \frac{\pi}{2} - \lim_{a \rightarrow 1^-} (1 - \sqrt{1-a^2}) = \frac{\pi}{2} - 1$$

$$7. (a) \int_1^\infty \frac{x}{\sqrt{1+x^5}} dx \text{ converges by comparison with } \int_1^\infty \frac{dx}{x^{3/2}}$$

$$(b) \int_1^\infty 2^{-x^2} dx = \int_1^\infty (e^{-x^2})^{\ln 2} dx \text{ converges by comparison with } \int_1^\infty e^{-x} dx$$

$$8. (a) \text{ The function } f \text{ is nonnegative on } (-\infty, \infty) \text{ and}$$

$$\int_{-\infty}^\infty f(x) dx = \int_{-\infty}^0 0 dx + \int_0^\infty \frac{6x}{(1+3x^2)^2} dx = \int_0^\infty \frac{6x}{(1+3x^2)^2} dx$$

$$\text{Now, } \int \frac{6x}{(1+3x^2)^2} dx = -\frac{1}{1+3x^2} + C$$

$$\text{Therefore, } \int_{-\infty}^\infty f(x) dx = \lim_{b \rightarrow \infty} \left[ -\frac{1}{1+3x^2} \right]_0^b = \lim_{b \rightarrow \infty} \left( 1 - \frac{1}{1+3b^2} \right) = 1$$

$$(b) f \text{ is nonnegative and } \int_{-\infty}^\infty f(x) dx = \int_0^\infty k e^{-kx} dx = \lim_{b \rightarrow \infty} [-e^{-kx}]_0^b = \lim_{b \rightarrow \infty} (-e^{-kb} + 1) = 1$$

So,  $f$  is a probability function.

$$(c) \mu = \int_{-\infty}^\infty x f(x) dx = \int_{-\infty}^0 0 dx + \int_0^\infty k x e^{-kx} dx = \lim_{b \rightarrow \infty} \int_0^b k x e^{-kx} dx$$

$$\text{Using integration by parts, } \int k x e^{-kx} dx = -x e^{-kx} - \frac{1}{k} e^{-kx} + C$$

Therefore,

$$\mu = \int_{-\infty}^\infty x f(x) dx = \lim_{b \rightarrow \infty} \left[ -x e^{-kx} - \frac{1}{k} e^{-kx} \right]_0^b = \lim_{b \rightarrow \infty} \left[ -b e^{-kb} - \frac{1}{k} e^{-kb} + \frac{1}{k} \right] = \frac{1}{k}$$

$$(d) \sigma = \int_{-\infty}^\infty (x - \mu)^2 f(x) dx = \int_0^\infty \left( x - \frac{1}{k} \right)^2 k e^{-kx} dx$$

$$= \int_0^\infty k x^2 e^{-kx} dx - 2 \int_0^\infty x e^{-kx} dx + \frac{1}{k} \int_0^\infty e^{-kx} dx = \frac{1}{k^2}$$

$$9. (a) \text{ Since } \lim_{n \rightarrow \infty} \frac{n}{\ln n} \text{ diverges, then } \sum (-1)^k \frac{k}{\ln k} \text{ diverges.}$$

$$(b) \text{ Let } a_n = \frac{n!}{2^n}$$

$$\text{Since } \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{n!}{2^n} \text{ diverges, then } \sum \frac{k!}{(-2)^k} = \sum (-1)^k \frac{k!}{2^k} \text{ diverges.}$$

$$(c) \text{ Since } \lim_{n \rightarrow \infty} \sin\left(\frac{n\pi}{4}\right) \text{ does not exist, then } \sum \sin\left(\frac{k\pi}{4}\right) \text{ diverges.}$$

$$(d) \text{ Let } a_n = \frac{(-1)^n}{n - 2\sqrt{n}} \Rightarrow |a_n| = \frac{1}{n - 2\sqrt{n}}, b_n = \frac{1}{n} \text{ and } \lim_{n \rightarrow \infty} \frac{|a_n|}{b_n} = 1$$

$$\text{Since } \sum b_n = \sum \frac{1}{n} \text{ diverges, then } \sum |a_n| \text{ diverges as well.}$$

And then  $\lim_{n \rightarrow \infty} \frac{1}{n - 2\sqrt{n}} = 0$  implies that  $\sum \frac{(-1)^k}{k - 2\sqrt{k}}$  converges.

Therefore,  $\sum \frac{(-1)^k}{k - 2\sqrt{k}}$  is conditionally convergent.

(e) Let  $a_n = (-1)^n \frac{n+2}{n^2+n} \Rightarrow |a_n| = \frac{n+2}{n^2+n}$ ,  $b_n = \frac{1}{n}$  and  $\lim_{n \rightarrow \infty} \frac{|a_n|}{b_n} = 1$

Since  $\sum b_n = \sum \frac{1}{n}$  diverges, then  $\sum |a_n|$  diverges as well.

And then  $\lim_{n \rightarrow \infty} \frac{n+2}{n^2+n} = 0$  implies that  $\sum (-1)^k \frac{k+2}{k^2+k}$  converges.

Therefore,  $\sum (-1)^k \frac{k+2}{k^2+k}$  is conditionally convergent.

(f) Since  $\lim_{n \rightarrow \infty} \frac{n^n}{n!}$  diverges, then  $\sum (-1)^{k+1} \frac{k^k}{k!}$  diverges.

(g) Let  $a_n = (-1)^n n e^{-n} \Rightarrow |a_n| = n e^{-n}$

Since  $\lim_{n \rightarrow \infty} \frac{|a_{n+1}|}{|a_n|} = \lim_{n \rightarrow \infty} \frac{n+1}{n e} = \frac{1}{e} < 1$ , then  $\sum |a_n|$  converges.

Therefore,  $\sum (-1)^k k e^{-k}$  is absolutely convergent.

(h) Let  $a_n = (-1)^n (\sqrt{n+1} - \sqrt{n}) \Rightarrow |a_n| = \sqrt{n+1} - \sqrt{n} = \frac{1}{\sqrt{n+1} + \sqrt{n}}$ ,  $b_n = \frac{1}{2\sqrt{k+1}}$

Since  $\sum b_n = \sum \frac{1}{2\sqrt{n+1}} < \sum |a_n| = \sum \frac{1}{\sqrt{n+1} + \sqrt{n}}$  and  $\sum \frac{1}{2\sqrt{n+1}}$  diverges,

then  $\sum |a_n| = \sum \frac{1}{\sqrt{k+1} + \sqrt{k}}$  diverges as well.

And then  $\lim_{n \rightarrow \infty} \sqrt{n+1} - \sqrt{n} = 0$  implies that  $\sum (-1)^k (\sqrt{k+1} - \sqrt{k})$  converges.

Therefore,  $\sum (-1)^k (\sqrt{k+1} - \sqrt{k})$  is conditionally convergent.

10. Let  $L = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!}$  and  $S_n = \sum_{k=0}^n \frac{(-1)^k}{k!}$

$$\text{Then } L - S_n = \sum_{k=n+1}^{\infty} \frac{(-1)^k}{k!}$$

To find the least value of  $n$  such that  $|S_n - L| < 0.001$

$$\text{Since } L - S_n = \frac{(-1)^{n+1}}{(n+1)!} + \frac{(-1)^{n+2}}{(n+2)!} + \frac{(-1)^{n+3}}{(n+3)!} + \dots$$

So  $n$  is the least value such that  $|\frac{(-1)^{n+1}}{(n+1)!}| < 0.001$

Therefore,  $\frac{1}{(n+1)!} < 0.001$  implies that  $n = 6$ .

11. (a) Since  $\sum \frac{1}{k}$  diverges, and  $\sum \frac{(\ln k)^2}{k} > \sum \frac{1}{k}$

Therefore,  $\frac{(\ln k)^2}{k}$  diverges.

$$(b) \sum \frac{2^k k!}{k^k} = \sum \frac{(2k)(2k-2)\dots 2}{k \cdot k \cdot \dots k} = 2 \sum \frac{(2k-2)(2k-6)\dots 2}{k \cdot k \cdot \dots k}$$

and then  $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \frac{2}{e} < 1$

- Therefore,  $\sum \frac{2^k k!}{k^k}$  converges.
- (c) Since  $\int_2^\infty \frac{1}{x} \left(\frac{1}{\ln x}\right)^{3/2} dx = \frac{2}{\sqrt{\ln 2}}$   
 Therefore,  $\sum \frac{1}{k} \left(\frac{1}{\ln k}\right)^{3/2}$  converges.
- (d)  $\frac{1}{2} + \frac{2}{3^2} + \frac{4}{4^3} + \frac{8}{5^4} + \dots = \sum_{k=0}^\infty \frac{2^{k-1}}{(k+1)^k} = \frac{1}{2} \sum_{k=0}^\infty \frac{2^k}{(k+1)^k}$   
 and  $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{2}{n+2} = 0$   
 Therefore,  $\frac{1}{2} + \frac{2}{3^2} + \frac{4}{4^3} + \frac{8}{5^4} + \dots$  converges.
- (e) Since  $\frac{1}{1+2 \ln k} > \frac{1}{1+2k}$  ( $\because \ln k < k$ ) and  $\sum \frac{1}{1+2k}$  diverges.  
 Therefore,  $\sum \frac{1}{1+2 \ln k}$  diverges.
- (f) Since  $\frac{2 + \sin k}{k^2} < \frac{3}{k^3}$  and  $\sum \frac{3}{k^3}$  converges.  
 Therefore,  $\sum \frac{2 + \sin k}{k^2}$  converges.
- (g) Let  $\sum a_n = \sum \frac{2n}{(2n)!}$   
 $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{1}{2n(2n+1)} = 0 < 1$   
 Therefore,  $\sum \frac{2k}{(2k)!}$  converges.
- (h) Since  $\int_1^\infty x e^{-x^2} = \frac{1}{2e}$   
 Therefore,  $\sum k e^{-k^2}$  converges.
- (i) Let  $\sum a_n = \sum \frac{2n!}{(2n)!}$   
 $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{1}{2(2n+1)} = 0 < 1$   
 Therefore,  $\sum \frac{2k!}{(2k)!}$  converges.
- (j) Since  $\lim_{n \rightarrow \infty} \left(\frac{n+1}{n}\right)^n = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e \neq 0$   
 Therefore,  $\sum \left(\frac{k+1}{k}\right)^k$  diverges.
- (k) Since  $\lim_{n \rightarrow \infty} \frac{(-5)^n}{4^{n+1}} = \lim_{n \rightarrow \infty} \frac{1}{4} \left(\frac{-5}{4}\right)^n \neq 0$   
 Therefore,  $\sum \frac{(-5)^k}{4^{k+1}}$  diverges.